

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

Date: 26-02-2026

Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name: ADITHYA KUMAR

Reg. no.: 23BCE7208

1. Use MongoDB to implement the following DB operations

1. Create a database called 'vehicles' and *write* a MongoDB query to select database as "vehicles".
2. Write a MongoDB query to display all the databases.
3. Create a collection called 'two_wheelers'. (use capping) and Create a collection called 'four_wheelers'.
4. Add 5 two-wheeler details to the collection named 'two_wheelers'. Each document consists of following fields as bike_name, model (gear or gearless), category (100cc, 125cc, 150cc, 200cc), colors_available (red, black, blue, sport red etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.
5. Add 5 four-wheeler details to the collection named 'four_wheelers'. Each document consists of following fields as vehicle_name, model (commercial or own), category (car, lorry, bus, mini truck, heavy truck, containers), variants (vxi, zxi, petrol, diesel etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.
6. Write a MongoDB query to display all documents available in two_wheelers and four_wheelers.
7. Write a MongoDB query to display only vehicle name and price in all the collection of the database
8. Write a MongoDB query to display two_wheelers from a particular company
9. Write a MongoDB query to display four_wheelers available in diesel variants
10. Write a MongoDB query to display vehicles name, category and manufacturer details whose rating is more than 5.

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The image displays two screenshots of a Windows 11 desktop environment, showing a MongoDB terminal window with the following commands and output:

```
test> use vehicles
switched to db vehicles
vehicles> show dbs
admin    40.00 KiB
config  60.00 KiB
local   40.00 KiB
vehicles>
| db.createCollection("two_wheelers", { capped: true, size: 10485760 })
{ ok: 1 }
vehicles> db.createCollection("four_wheelers")
{ ok: 1 }
vehicles> db.two_wheelers.insertMany([
  {
    bike_name: "Pulsar 150",
    model: "gear",
    category: "150cc",
    colors_available: ["black", "blue", "red"],
    manufacturer: "Bajaj",
    performance: 7.5,
    timestamp: new Date("2023-01-15T00:00:00Z"),
    price: 112000
  },
  {
    bike_name: "Activa 6G",
    model: "gearless",
    category: "125cc",
    colors_available: ["red", "grey", "white"],
    manufacturer: "Honda",
    performance: 8.0,
  }
])
```

The second screenshot shows the insertion of a third document into the 'two_wheelers' collection:

```
vehicles> db.two_wheelers.insertMany([
  {
    bike_name: "Pulsar 150",
    model: "gear",
    category: "150cc",
    colors_available: ["black", "blue", "red"],
    manufacturer: "Bajaj",
    performance: 7.5,
    timestamp: new Date("2023-01-15T00:00:00Z"),
    price: 112000
  },
  {
    bike_name: "Activa 6G",
    model: "gearless",
    category: "125cc",
    colors_available: ["red", "grey", "white"],
    manufacturer: "Honda",
    performance: 8.0,
  },
  {
    bike_name: "Splendor",
    model: "gearless",
    category: "125cc",
    colors_available: ["red", "grey", "white"],
    manufacturer: "Honda",
    performance: 8.0,
    timestamp: new Date("2022-03-10T00:00:00Z"),
    price: 85000
  }
])
```

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```
category: "125cc",
colors_available: ["red", "grey", "white"],
manufacturer: "Honda",
performance: 8.0,
timestamp: new Date("2022-03-10T00:00:00Z"),
price: 85000
},
{
  bike_name: "Splendor+",
  model: "gear",
  category: "100cc",
  colors_available: ["black", "red", "silver"],
  manufacturer: "Hero",
  performance: 6.8,
  timestamp: new Date("2021-07-22T00:00:00Z"),
  price: 75000
},
{
  bike_name: "Royal Enfield Classic 350",
  model: "gear",
  category: "350cc", // Added 350cc as it fits the data
  colors_available: ["stealth black", "chrome red", "halcyon green"],
  manufacturer: "Royal Enfield",
  performance: 7.9,
  timestamp: new Date("2021-09-01T00:00:00Z"),
  price: 190000
},
{
  bike_name: "R15 V4",
  model: "gear",
  category: "150cc",
  colors_available: ["racing blue", "metallic grey", "dark knight"],
  manufacturer: "Yamaha",
  performance: 8.5,
  timestamp: new Date("2022-04-12T00:00:00Z"),
  price: 185000
}
];
acknowledged: true,
insertedIds: {
  0: ObjectId('69a0760647788770187c2907'),
  1: ObjectId('69a0760647788770187c2908'),
  2: ObjectId('69a0760647788770187c2909'),
  3: ObjectId('69a0760647788770187c290a'),
  4: ObjectId('69a0760647788770187c290b')
}
}
vehicles>
(To exit, press Ctrl+C again or Ctrl+D or type .exit)
vehicles> db.four_wheelers.insertMany([
  {
    vehicle_name: "Innova Crysta",
    model: "commercial",
    category: "car",
    variants: ["vx", "zxi", "diesel"],
    manufacturer: "Toyota",
    performance: 8.2,
    timestamp: new Date("2020-11-05T00:00:00Z"),
```

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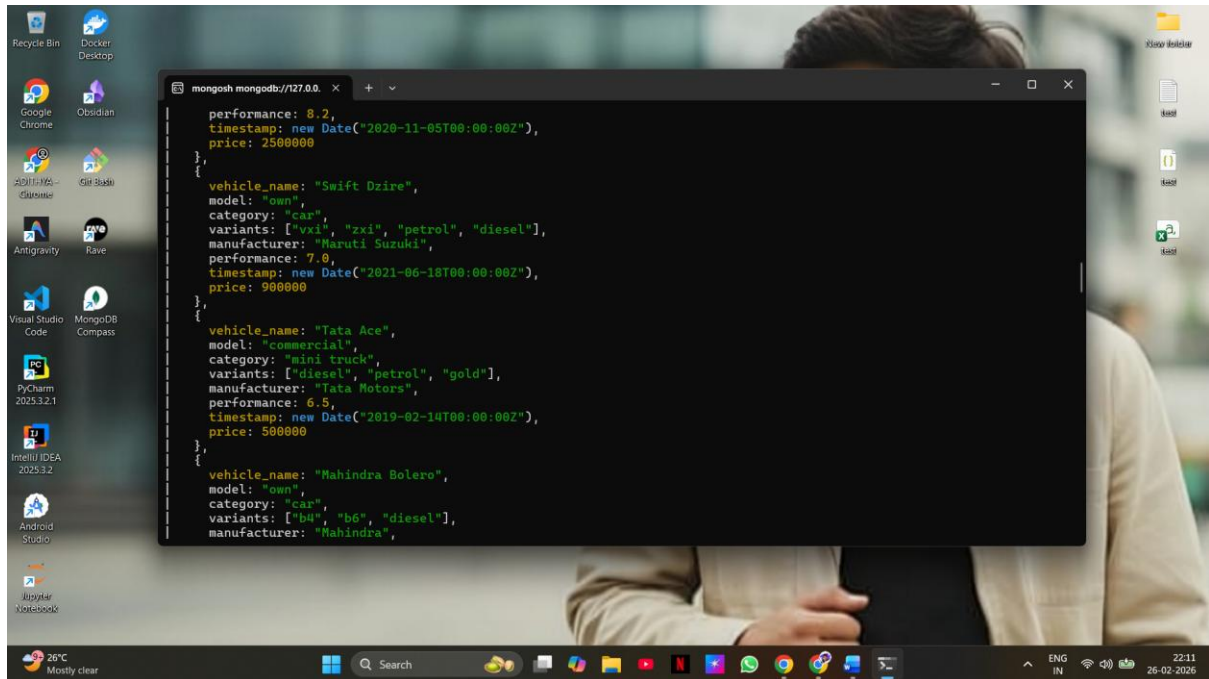
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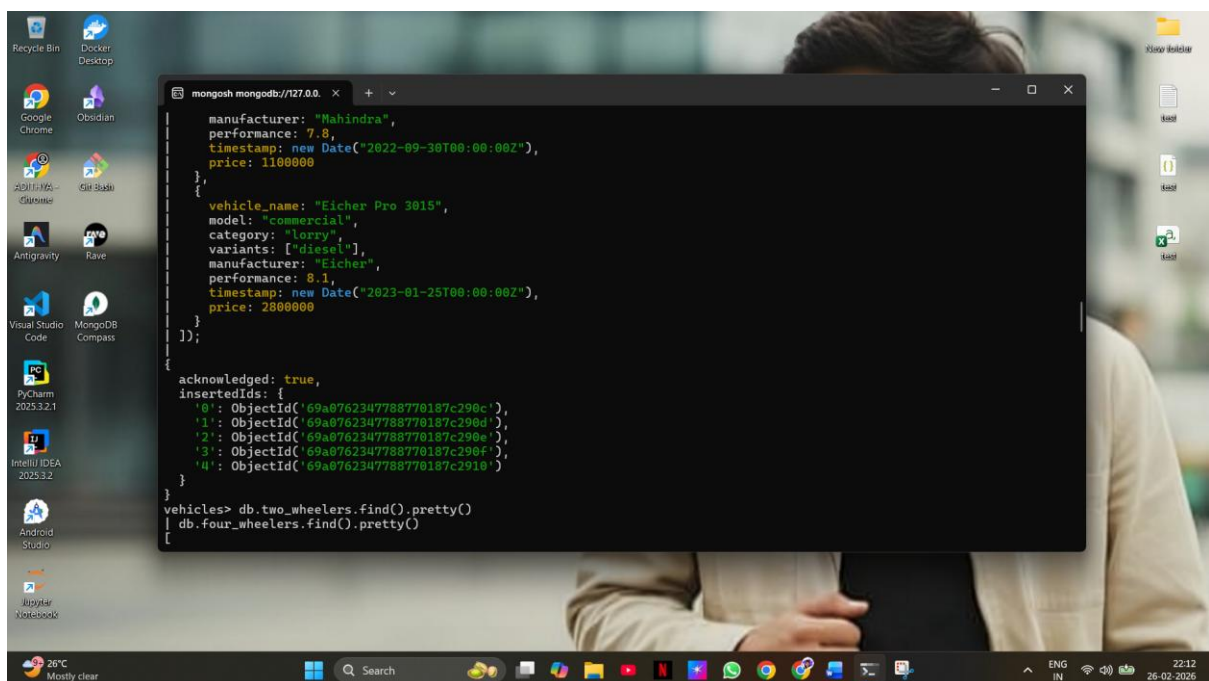
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```
mongosh mongodb://127.0.0.1:27000
{
  performance: 8.2,
  timestamp: new Date("2020-11-05T00:00:00Z"),
  price: 2500000
},
{
  vehicle_name: "Swift Dzire",
  model: "own",
  category: "car",
  variants: ["vxi", "zxi", "petrol", "diesel"],
  manufacturer: "Maruti Suzuki",
  performance: 7.0,
  timestamp: new Date("2021-06-18T00:00:00Z"),
  price: 900000
},
{
  vehicle_name: "Tata Ace",
  model: "commercial",
  category: "mini truck",
  variants: ["diesel", "petrol", "gold"],
  manufacturer: "Tata Motors",
  performance: 6.5,
  timestamp: new Date("2019-02-14T00:00:00Z"),
  price: 500000
},
{
  vehicle_name: "Mahindra Bolero",
  model: "own",
  category: "car",
  variants: ["b6", "b6", "diesel"],
  manufacturer: "Mahindra",
  price: 1100000
}
```



```
mongosh mongodb://127.0.0.1:27000
{
  manufacturer: "Mahindra",
  performance: 7.8,
  timestamp: new Date("2022-09-30T00:00:00Z"),
  price: 1100000
},
{
  vehicle_name: "Eicher Pro 3015",
  model: "commercial",
  category: "lorry",
  variants: ["diesel"],
  manufacturer: "Eicher",
  performance: 8.1,
  timestamp: new Date("2023-01-25T00:00:00Z"),
  price: 2800000
}
}

acknowledged: true,
insertedIds: {
  '0': ObjectId('69a0762347788770187c290c'),
  '1': ObjectId('69a0762347788770187c290d'),
  '2': ObjectId('69a0762347788770187c290e'),
  '3': ObjectId('69a0762347788770187c290f'),
  '4': ObjectId('69a0762347788770187c2910')
}
vehicles> db.two_wheelers.find().pretty()
| db.four_wheelers.find().pretty()
[
```


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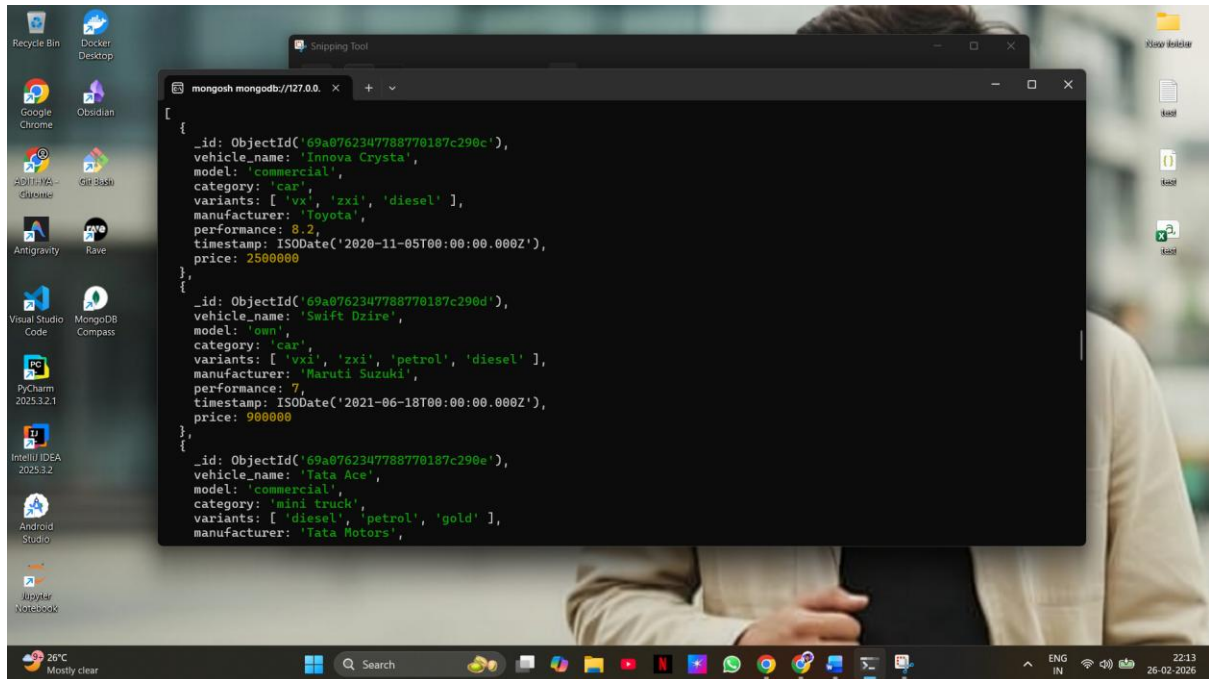
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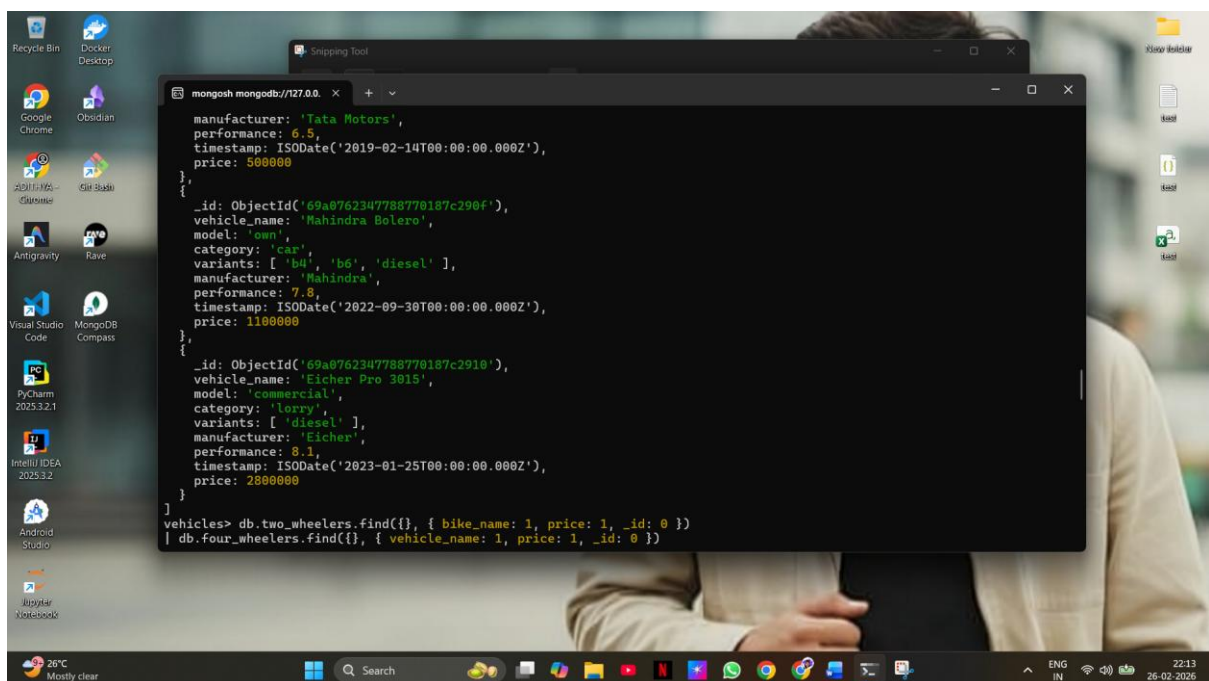
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The screenshot shows a Windows desktop with a MongoDB terminal window open. The terminal displays a list of vehicles with their details, including _id, vehicle_name, model, category, variants, manufacturer, performance, timestamp, and price. The vehicles listed are Innova Crysta, Swift Dzire, and Tata Ace.

```
[ {
  _id: ObjectId('69a0762347788770187c290c'),
  vehicle_name: 'Innova Crysta',
  model: 'commercial',
  category: 'car',
  variants: [ 'vx', 'zx1', 'diesel' ],
  manufacturer: 'Toyota',
  performance: 8.2,
  timestamp: ISODate('2020-11-05T00:00:00.000Z'),
  price: 2500000
}, {
  _id: ObjectId('69a0762347788770187c290d'),
  vehicle_name: 'Swift Dzire',
  model: 'own',
  category: 'car',
  variants: [ 'vxi', 'zx1', 'petrol', 'diesel' ],
  manufacturer: 'Maruti Suzuki',
  performance: 7,
  timestamp: ISODate('2021-06-18T00:00:00.000Z'),
  price: 900000
}, {
  _id: ObjectId('69a0762347788770187c290e'),
  vehicle_name: 'Tata Ace',
  model: 'commercial',
  category: 'mini truck',
  variants: [ 'diesel', 'petrol', 'gold' ],
  manufacturer: 'Tata Motors',
  performance: 6.5,
  timestamp: ISODate('2019-02-14T00:00:00.000Z'),
  price: 500000
}, {
  _id: ObjectId('69a0762347788770187c290f'),
  vehicle_name: 'Mahindra Bolero',
  model: 'own',
  category: 'car',
  variants: [ 'b4', 'b6', 'diesel' ],
  manufacturer: 'Mahindra',
  performance: 7.8,
  timestamp: ISODate('2022-09-30T00:00:00.000Z'),
  price: 1100000
}, {
  _id: ObjectId('69a0762347788770187c2910'),
  vehicle_name: 'Eicher Pro 3015',
  model: 'commercial',
  category: 'lorry',
  variants: [ 'diesel' ],
  manufacturer: 'Eicher',
  performance: 8.1,
  timestamp: ISODate('2023-01-25T00:00:00.000Z'),
  price: 2800000
} ]
vehicles> db.two_wheelers.find({}, { bike_name: 1, price: 1, _id: 0 })
| db.four_wheelers.find({}, { vehicle_name: 1, price: 1, _id: 0 })
```



The screenshot shows a Windows desktop with a MongoDB terminal window open. The terminal displays a list of vehicles with their details, including _id, vehicle_name, model, category, variants, manufacturer, performance, timestamp, and price. The vehicles listed are Tata Ace, Mahindra Bolero, Eicher Pro 3015, and Mahindra Bolero. The terminal also shows the output of the find command for two-wheelers and four-wheelers.

```
manufacturer: 'Tata Motors',
performance: 6.5,
timestamp: ISODate('2019-02-14T00:00:00.000Z'),
price: 500000
}, {
  _id: ObjectId('69a0762347788770187c290f'),
  vehicle_name: 'Mahindra Bolero',
  model: 'own',
  category: 'car',
  variants: [ 'b4', 'b6', 'diesel' ],
  manufacturer: 'Mahindra',
  performance: 7.8,
  timestamp: ISODate('2022-09-30T00:00:00.000Z'),
  price: 1100000
}, {
  _id: ObjectId('69a0762347788770187c2910'),
  vehicle_name: 'Eicher Pro 3015',
  model: 'commercial',
  category: 'lorry',
  variants: [ 'diesel' ],
  manufacturer: 'Eicher',
  performance: 8.1,
  timestamp: ISODate('2023-01-25T00:00:00.000Z'),
  price: 2800000
} ]
vehicles> db.two_wheelers.find({}, { bike_name: 1, price: 1, _id: 0 })
| db.four_wheelers.find({}, { vehicle_name: 1, price: 1, _id: 0 })
```

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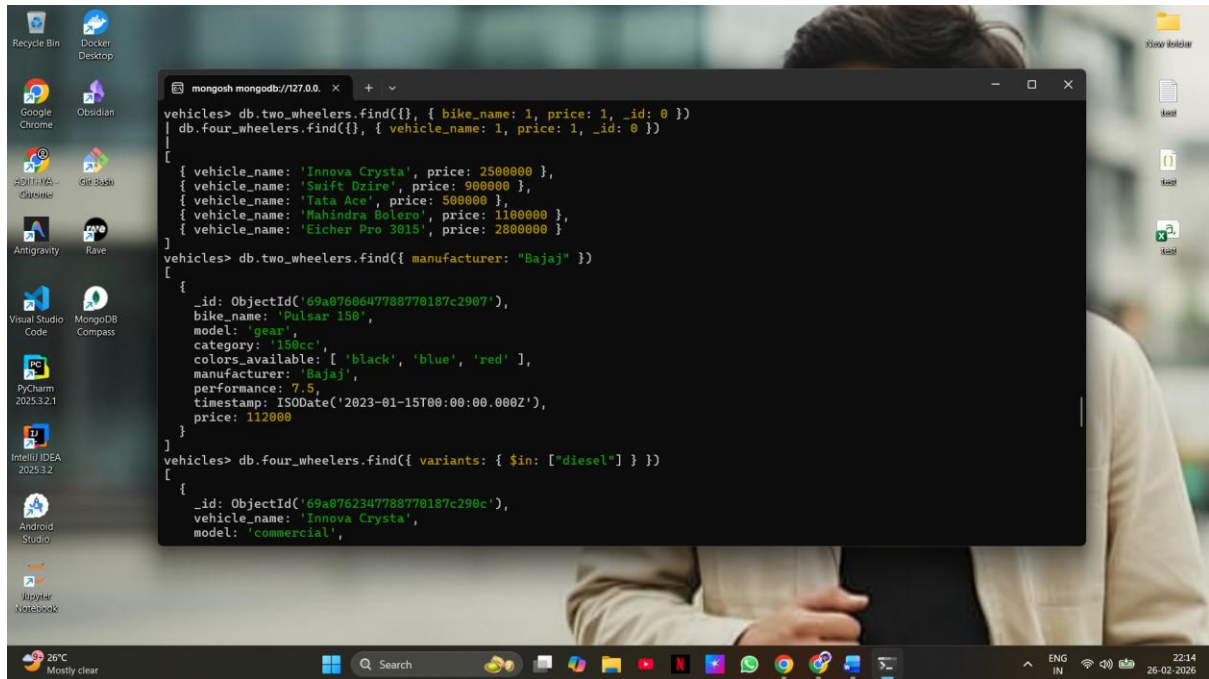
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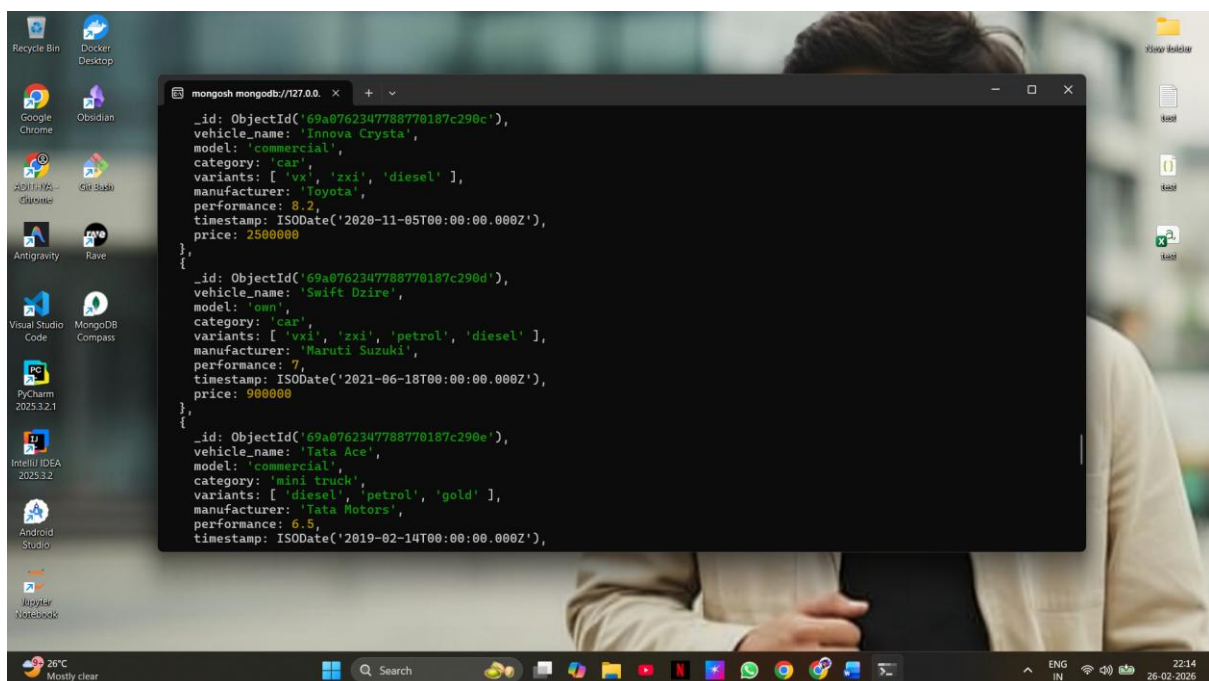
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```
vehicles> db.two_wheelers.find({}, { bike_name: 1, price: 1, _id: 0 })
| db.four_wheelers.find({}, { vehicle_name: 1, price: 1, _id: 0 })
[
  { vehicle_name: 'Innova Crysta', price: 2500000 },
  { vehicle_name: 'Swift Dzire', price: 900000 },
  { vehicle_name: 'Tata Ace', price: 500000 },
  { vehicle_name: 'Mahindra Bolero', price: 1100000 },
  { vehicle_name: 'Eicher Pro 3015', price: 2000000 }
]
vehicles> db.two_wheelers.find({ manufacturer: "Bajaj" })
[
  {
    _id: ObjectId('69a0760647788770187c2907'),
    bike_name: 'Pulsar 150',
    model: 'gear',
    category: '150cc',
    colors_available: [ 'black', 'blue', 'red' ],
    manufacturer: 'Bajaj',
    performance: 7.5,
    timestamp: ISODate('2023-01-15T00:00:00.000Z'),
    price: 112000
  }
]
vehicles> db.four_wheelers.find({ variants: { $in: ["diesel"] } })
[
  {
    _id: ObjectId('69a0762347788770187c290c'),
    vehicle_name: 'Innova Crysta',
    model: 'commercial',
    category: 'car',
    variants: [ 'vx', 'zx', 'diesel' ],
    manufacturer: 'Toyota',
    performance: 8.2,
    timestamp: ISODate('2020-11-05T00:00:00.000Z'),
    price: 2500000
  },
  {
    _id: ObjectId('69a0762347788770187c290d'),
    vehicle_name: 'Swift Dzire',
    model: 'own',
    category: 'car',
    variants: [ 'vxi', 'zx', 'petrol', 'diesel' ],
    manufacturer: 'Maruti Suzuki',
    performance: 7,
    timestamp: ISODate('2021-06-18T00:00:00.000Z'),
    price: 900000
  },
  {
    _id: ObjectId('69a0762347788770187c290e'),
    vehicle_name: 'Tata Ace',
    model: 'commercial',
    category: 'mini truck',
    variants: [ 'diesel', 'petrol', 'gold' ],
    manufacturer: 'Tata Motors',
    performance: 6.5,
    timestamp: ISODate('2019-02-14T00:00:00.000Z'),
    price: 500000
  }
]
```



```
{
  _id: ObjectId('69a0762347788770187c290c'),
  vehicle_name: 'Innova Crysta',
  model: 'commercial',
  category: 'car',
  variants: [ 'vx', 'zx', 'diesel' ],
  manufacturer: 'Toyota',
  performance: 8.2,
  timestamp: ISODate('2020-11-05T00:00:00.000Z'),
  price: 2500000
},
{
  _id: ObjectId('69a0762347788770187c290d'),
  vehicle_name: 'Swift Dzire',
  model: 'own',
  category: 'car',
  variants: [ 'vxi', 'zx', 'petrol', 'diesel' ],
  manufacturer: 'Maruti Suzuki',
  performance: 7,
  timestamp: ISODate('2021-06-18T00:00:00.000Z'),
  price: 900000
},
{
  _id: ObjectId('69a0762347788770187c290e'),
  vehicle_name: 'Tata Ace',
  model: 'commercial',
  category: 'mini truck',
  variants: [ 'diesel', 'petrol', 'gold' ],
  manufacturer: 'Tata Motors',
  performance: 6.5,
  timestamp: ISODate('2019-02-14T00:00:00.000Z'),
  price: 500000
}
]
```

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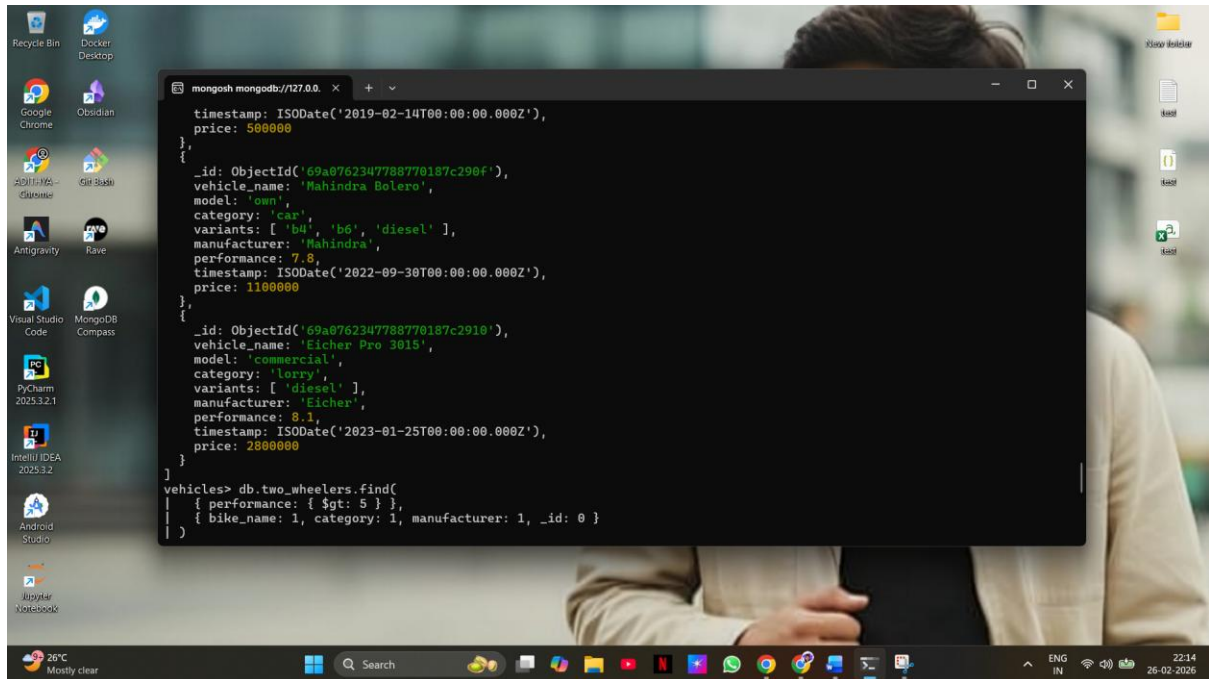
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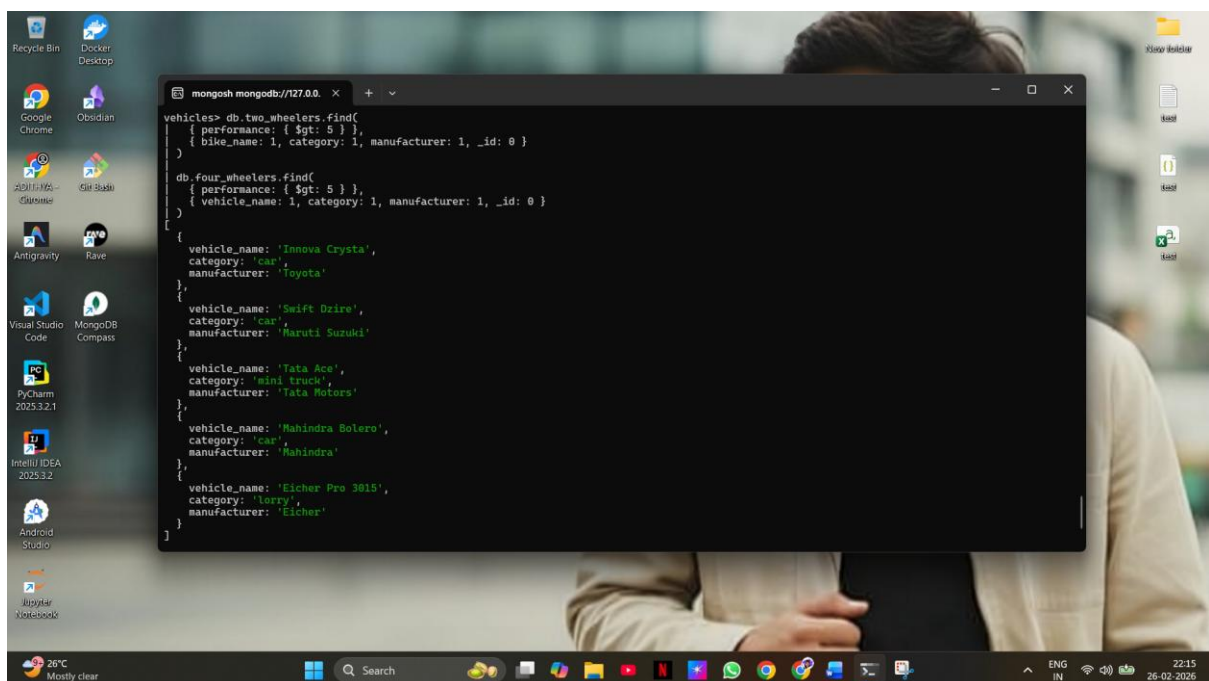
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Reg. no.: 23BCE7208



```
timestamp: ISODate('2019-02-14T00:00:00.000Z'),
price: 500000
},
{
  _id: ObjectId('69a0762347788770187c290f'),
  vehicle_name: 'Mahindra Bolero',
  model: 'own',
  category: 'car',
  variants: [ 'b0', 'b6', 'diesel' ],
  manufacturer: 'Mahindra',
  performance: 7.8,
  timestamp: ISODate('2022-09-30T00:00:00.000Z'),
  price: 1100000
},
{
  _id: ObjectId('69a0762347788770187c2910'),
  vehicle_name: 'Eicher Pro 3015',
  model: 'commercial',
  category: 'lorry',
  variants: [ 'diesel' ],
  manufacturer: 'Eicher',
  performance: 8.1,
  timestamp: ISODate('2023-01-25T00:00:00.000Z'),
  price: 2000000
}
]
vehicles> db.two_wheelers.find(
  { performance: { $gt: 5 } },
  { bike_name: 1, category: 1, manufacturer: 1, _id: 0 }
)
```



```
vehicles> db.two_wheelers.find(
  { performance: { $gt: 5 } },
  { bike_name: 1, category: 1, manufacturer: 1, _id: 0 }
)
db.four_wheelers.find(
  { performance: { $gt: 5 } },
  { vehicle_name: 1, category: 1, manufacturer: 1, _id: 0 }
)
[
  {
    vehicle_name: 'Innova Crysta',
    category: 'car',
    manufacturer: 'Toyota'
  },
  {
    vehicle_name: 'Swift Dzire',
    category: 'car',
    manufacturer: 'Maruti Suzuki'
  },
  {
    vehicle_name: 'Tata Ace',
    category: 'mini truck',
    manufacturer: 'Tata Motors'
  },
  {
    vehicle_name: 'Mahindra Bolero',
    category: 'car',
    manufacturer: 'Mahindra'
  },
  {
    vehicle_name: 'Eicher Pro 3015',
    category: 'lorry',
    manufacturer: 'Eicher'
  }
]
```

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2. Use MongoDB to implement the following DB operations for a Zoo

1. Create a database called 'animal' and *write* a MongoDB query to select database as 'animal'.
2. Write a MongoDB query to display all the databases.
3. Create a collection called 'wild_animals'.(use capping) and Create a collection called 'domestic_animals'.
4. Add 5 wild_animal details to the collection named 'wild_animals'. Each document consists of following fields as animal_name, nature (harm or harmless), favorite_foods (meat, rabbits, deer etc) as array, care_taker_name, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.
5. Add 5 domestic-animal details to the collection named 'domestic_animals'. Each document consists of following fields as animal_name, gender (male or female), favorite_foods (meat, rabbits, deer etc) as array, animal_petname, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.
6. Write a MongoDB query to display all documents available in wild_animals and domestic_animals.
7. Write a MongoDB query to display only animal name and expenses in all the collection of the database
8. Write a MongoDB query to display domestic_animals whose life is a particular year
9. Write a MongoDB query to display wild_animals available under a particular care_taker
10. Write a MongoDB query to display animal name, favorite_foods and expenses details whose lifespan is more than 5 years.

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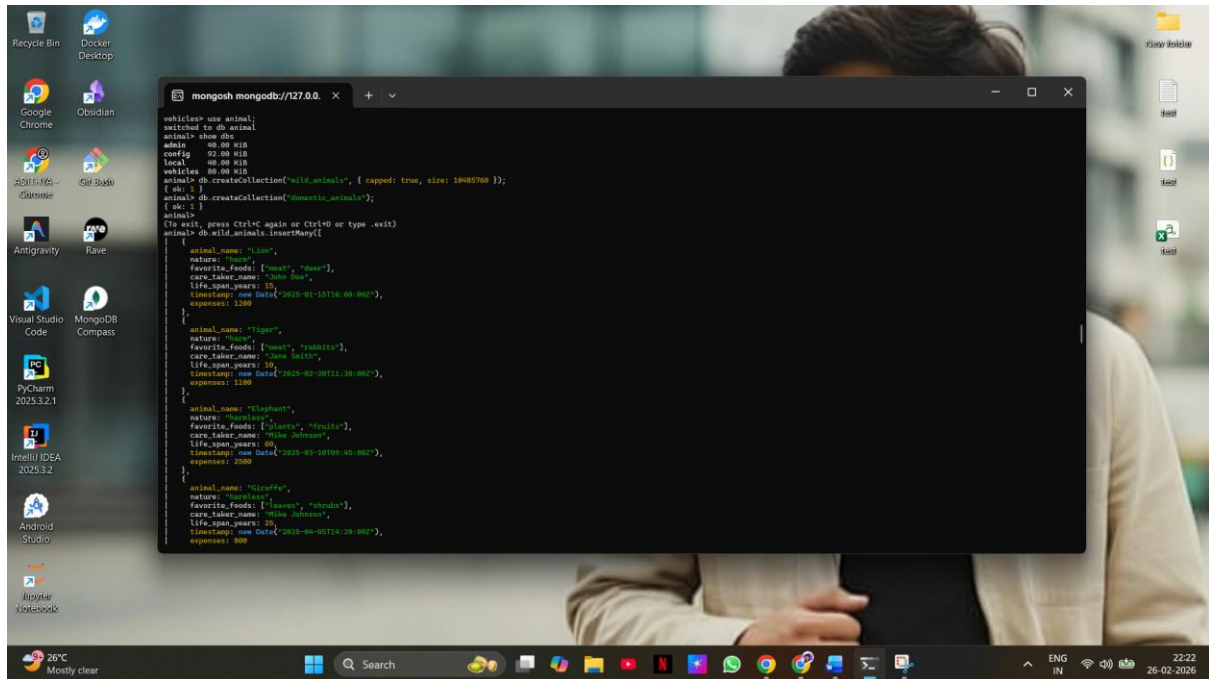
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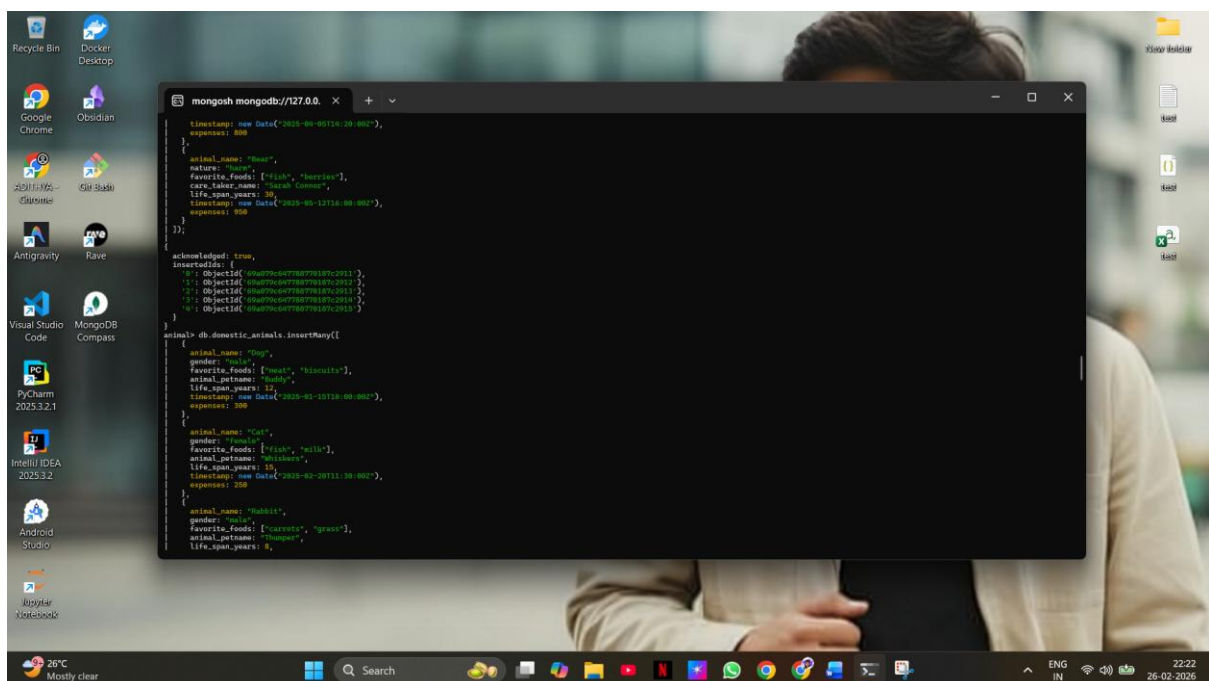
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```
vehicles> use animal
switched to db animal
animals> show dbs
admin    48.00 MiB
config  92.00 MiB
local   48.00 MiB
vehicles 88.00 MiB
animals> db.createCollection('wild_animals', { capped: true, size: 10000000 });
{ ok: 1 }
animals> db.createCollection('domestic_animals');
{ ok: 1 }
animals>
It's exit, press Ctrl+C again or Ctrl+D or type .exit
animals> db.wild_animals.insertMany([
  {
    animal_name: "Lion",
    nature: "lion",
    favorite_foods: ["meat", "deer"],
    caretaker_name: "John Doe",
    life_span_years: 15,
    timestamp: new Date("2025-01-15T18:00:00Z"),
    expenses: 1200
  },
  {
    animal_name: "Tiger",
    nature: "lion",
    favorite_foods: ["meat", "rabbits"],
    caretaker_name: "John Doe",
    life_span_years: 10,
    timestamp: new Date("2025-02-20T11:30:00Z"),
    expenses: 1100
  },
  {
    animal_name: "Elephant",
    nature: "herbivore",
    favorite_foods: ["plants", "fruits"],
    caretaker_name: "Mike Johnson",
    life_span_years: 60,
    timestamp: new Date("2025-03-10T00:40:00Z"),
    expenses: 2500
  },
  {
    animal_name: "Giraffe",
    nature: "herbivore",
    favorite_foods: ["leaves", "shrubs"],
    caretaker_name: "Mike Johnson",
    life_span_years: 25,
    timestamp: new Date("2025-04-05T14:30:00Z"),
    expenses: 800
  }
])
```



```
timestamp: new Date("2025-04-05T14:30:00Z"),
expenses: 800
},
{
  animal_name: "Bear",
  nature: "lion",
  favorite_foods: ["fish", "berries"],
  caretaker_name: "Sarah Connor",
  life_span_years: 30,
  timestamp: new Date("2025-05-12T15:00:00Z"),
  expenses: 950
}
]),
acknowledged: true,
insertedIds: [
  ObjectId("60a079c0d778b778187c2911"),
  ObjectId("60a079c0d778b778187c2912"),
  ObjectId("60a079c0d778b778187c2913"),
  ObjectId("60a079c0d778b778187c2914"),
  ObjectId("60a079c0d778b778187c2915")
]
}
animals> db.domestic_animals.insertMany([
  {
    animal_name: "Dog",
    gender: "male",
    favorite_foods: ["meat", "biscuits"],
    animal_name: "Bully",
    life_span_years: 12,
    timestamp: new Date("2025-03-10T18:00:00Z"),
    expenses: 300
  },
  {
    animal_name: "Cat",
    gender: "female",
    favorite_foods: ["fish", "milk"],
    animal_name: "Whiskers",
    life_span_years: 15,
    timestamp: new Date("2025-03-20T11:30:00Z"),
    expenses: 250
  },
  {
    animal_name: "Rabbit",
    gender: "male",
    favorite_foods: ["carrots", "grass"],
    animal_name: "Thumper",
    life_span_years: 8
  }
])
```

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

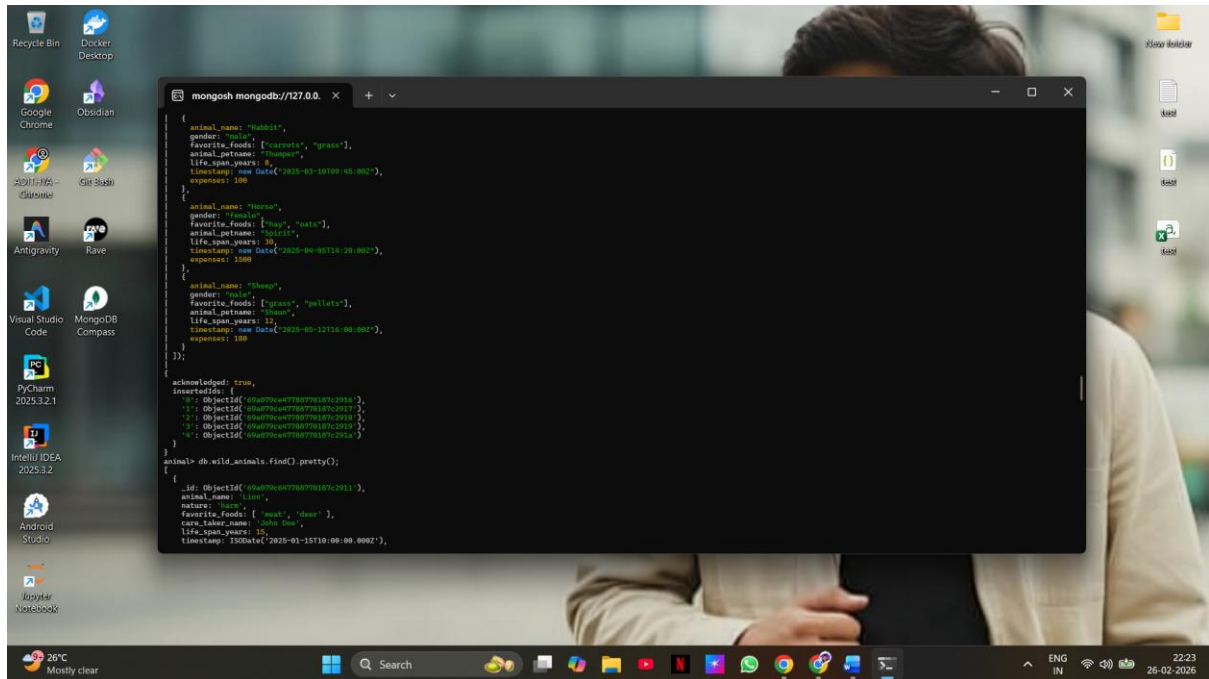
Date: 26-02-2026

Faculty Name: Prof. S.Gopikrishnan

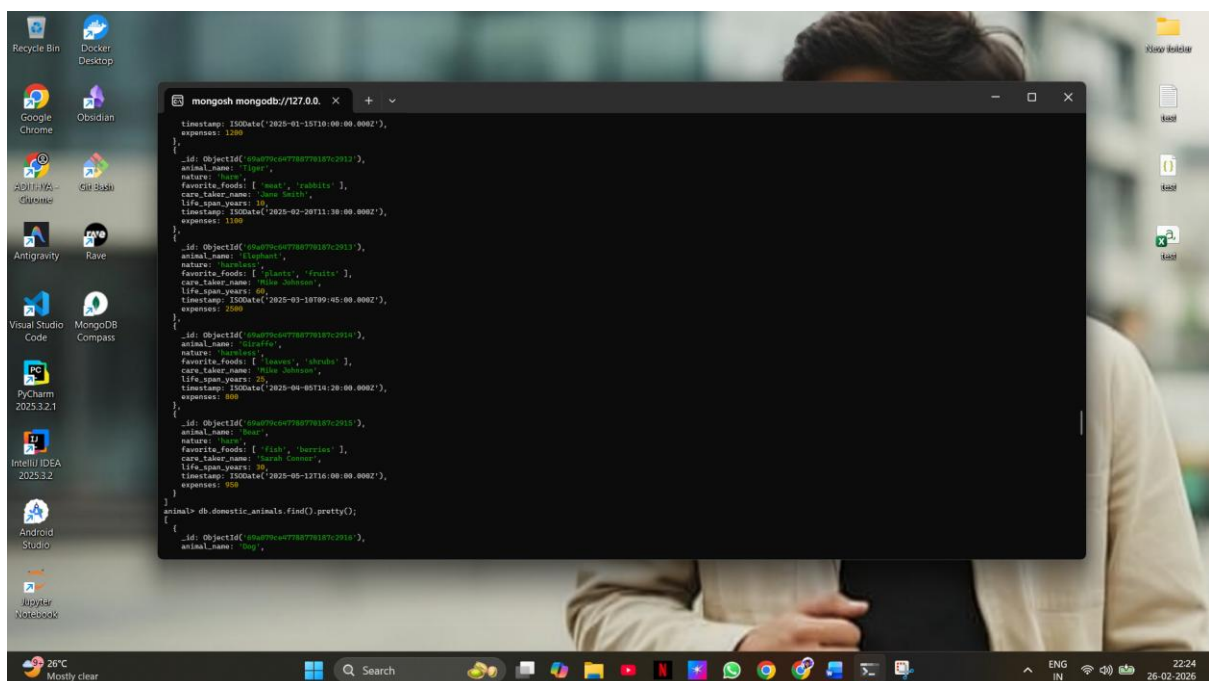
School: SCOPE

Student name: ADITHYA KUMAR

Reg. no.: 23BCE7208



```
mongosh mongodb://127.0.0.1:27020
> use test
> use animals
> insertMany([
  {
    animal_name: "Rabbit",
    gender: "male",
    favorite_foods: ["carrots", "grass"],
    animal_potname: "Thumper",
    life_span_years: 8,
    timestamp: new Date("2025-03-10T09:45.000Z"),
    expenses: 100
  },
  {
    animal_name: "Horse",
    gender: "female",
    favorite_foods: ["hay", "mats"],
    animal_potname: "Spirrit",
    life_span_years: 30,
    timestamp: new Date("2025-04-05T10:20.000Z"),
    expenses: 1500
  },
  {
    animal_name: "Sheep",
    gender: "male",
    favorite_foods: ["grass", "pellets"],
    animal_potname: "Sham",
    life_span_years: 12,
    timestamp: new Date("2025-05-12T10:00.000Z"),
    expenses: 100
  }
]);
acknowledged: true,
insertedIds: [
  ObjectId('64a079cc4778b778187c291a'),
  ObjectId('64a079cc4778b778187c2917'),
  ObjectId('64a079cc4778b778187c2918')
]
> animals.db.wild_animals.find().pretty();
{
  "_id": ObjectId("64a079cc4778b778187c2911"),
  "animal_name": "lion",
  "nature": "lion",
  "favorite_foods": [ "meat", "deer" ],
  "care_taker_name": "John Doe",
  "life_span_years": 15,
  "timestamp": ISODate("2025-01-15T10:00:00.000Z"),
  "expenses": 1200
},
{
  "_id": ObjectId("64a079cc4778b778187c2912"),
  "animal_name": "tiger",
  "nature": "lion",
  "favorite_foods": [ "meat", "rodents" ],
  "care_taker_name": "John Smith",
  "life_span_years": 30,
  "timestamp": ISODate("2025-02-20T11:30:00.000Z"),
  "expenses": 1100
},
{
  "_id": ObjectId("64a079cc4778b778187c2913"),
  "animal_name": "elephant",
  "nature": "herbivore",
  "favorite_foods": [ "plants", "fruits" ],
  "care_taker_name": "Mike Johnson",
  "life_span_years": 60,
  "timestamp": ISODate("2025-03-10T09:45:00.000Z"),
  "expenses": 2500
},
{
  "_id": ObjectId("64a079cc4778b778187c2914"),
  "animal_name": "giraffe",
  "nature": "herbivore",
  "favorite_foods": [ "leaves", "shrubs" ],
  "care_taker_name": "Mike Johnson",
  "life_span_years": 20,
  "timestamp": ISODate("2025-04-05T14:20:00.000Z"),
  "expenses": 800
},
{
  "_id": ObjectId("64a079cc4778b778187c2915"),
  "animal_name": "bear",
  "nature": "herb",
  "favorite_foods": [ "fish", "berries" ],
  "care_taker_name": "Sarah Connor",
  "life_span_years": 35,
  "timestamp": ISODate("2025-05-12T16:00:00.000Z"),
  "expenses": 950
}
> animals.db.domestic_animals.find().pretty();
{
  "_id": ObjectId("64a079cc4778b778187c2916"),
  "animal_name": "dog",
  "nature": "herbivore",
  "favorite_foods": [ "meat", "vegetables" ],
  "care_taker_name": "John Doe",
  "life_span_years": 15,
  "timestamp": ISODate("2025-01-15T10:00:00.000Z"),
  "expenses": 1200
},
{
  "_id": ObjectId("64a079cc4778b778187c2917"),
  "animal_name": "cat",
  "nature": "herbivore",
  "favorite_foods": [ "fish", "birds" ],
  "care_taker_name": "John Smith",
  "life_span_years": 20,
  "timestamp": ISODate("2025-02-20T11:30:00.000Z"),
  "expenses": 1100
},
{
  "_id": ObjectId("64a079cc4778b778187c2918"),
  "animal_name": "parrot",
  "nature": "herbivore",
  "favorite_foods": [ "fruits", "seeds" ],
  "care_taker_name": "Mike Johnson",
  "life_span_years": 50,
  "timestamp": ISODate("2025-03-10T09:45:00.000Z"),
  "expenses": 2500
},
{
  "_id": ObjectId("64a079cc4778b778187c2919"),
  "animal_name": "hamster",
  "nature": "herbivore",
  "favorite_foods": [ "grains", "vegetables" ],
  "care_taker_name": "Mike Johnson",
  "life_span_years": 5,
  "timestamp": ISODate("2025-04-05T14:20:00.000Z"),
  "expenses": 800
},
{
  "_id": ObjectId("64a079cc4778b778187c291a"),
  "animal_name": "goldfish",
  "nature": "herbivore",
  "favorite_foods": [ "fish", "algae" ],
  "care_taker_name": "Sarah Connor",
  "life_span_years": 10,
  "timestamp": ISODate("2025-05-12T16:00:00.000Z"),
  "expenses": 950
}
```

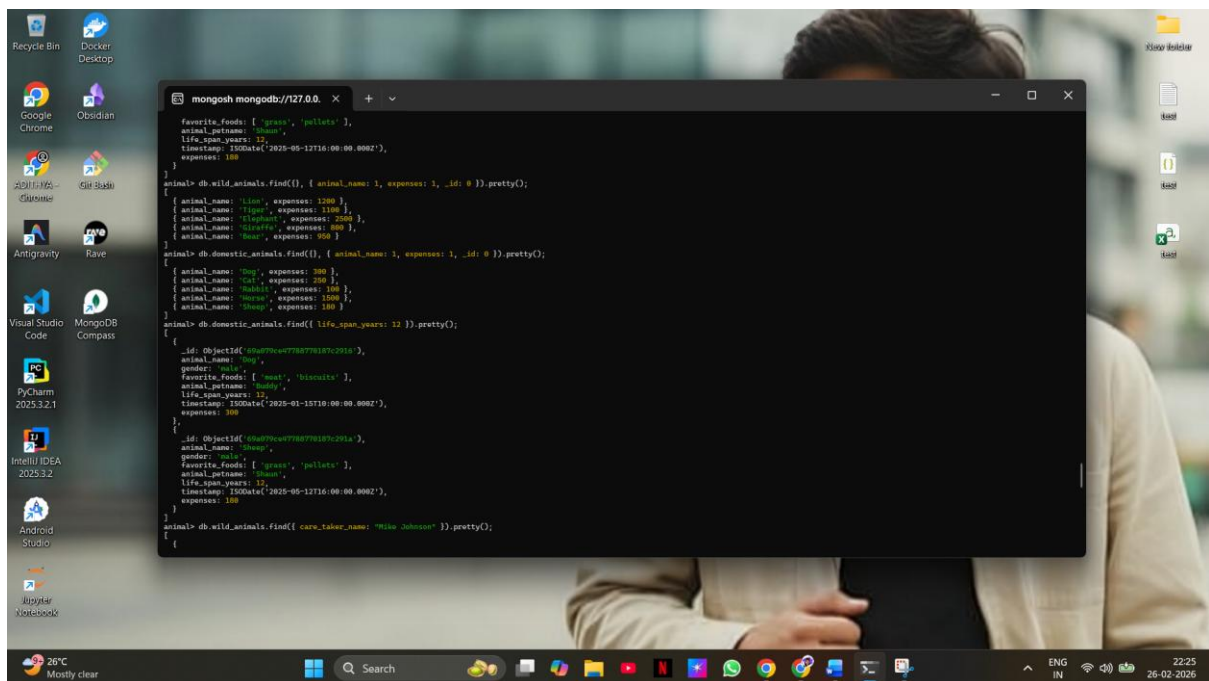


```
mongosh mongodb://127.0.0.1:27020
> use test
> use animals
> insertMany([
  {
    animal_name: "Rabbit",
    gender: "male",
    favorite_foods: ["carrots", "grass"],
    animal_potname: "Thumper",
    life_span_years: 8,
    timestamp: new Date("2025-03-10T09:45.000Z"),
    expenses: 100
  },
  {
    animal_name: "Horse",
    gender: "female",
    favorite_foods: ["hay", "mats"],
    animal_potname: "Spirrit",
    life_span_years: 30,
    timestamp: new Date("2025-04-05T10:20.000Z"),
    expenses: 1500
  },
  {
    animal_name: "Sheep",
    gender: "male",
    favorite_foods: ["grass", "pellets"],
    animal_potname: "Sham",
    life_span_years: 12,
    timestamp: new Date("2025-05-12T10:00.000Z"),
    expenses: 100
  }
]);
acknowledged: true,
insertedIds: [
  ObjectId('64a079cc4778b778187c291a'),
  ObjectId('64a079cc4778b778187c2917'),
  ObjectId('64a079cc4778b778187c2918')
]
> animals.db.wild_animals.find().pretty();
{
  "_id": ObjectId("64a079cc4778b778187c2911"),
  "animal_name": "lion",
  "nature": "lion",
  "favorite_foods": [ "meat", "deer" ],
  "care_taker_name": "John Doe",
  "life_span_years": 15,
  "timestamp": ISODate("2025-01-15T10:00:00.000Z"),
  "expenses": 1200
},
{
  "_id": ObjectId("64a079cc4778b778187c2912"),
  "animal_name": "tiger",
  "nature": "lion",
  "favorite_foods": [ "meat", "rodents" ],
  "care_taker_name": "John Smith",
  "life_span_years": 30,
  "timestamp": ISODate("2025-02-20T11:30:00.000Z"),
  "expenses": 1100
},
{
  "_id": ObjectId("64a079cc4778b778187c2913"),
  "animal_name": "elephant",
  "nature": "herbivore",
  "favorite_foods": [ "plants", "fruits" ],
  "care_taker_name": "Mike Johnson",
  "life_span_years": 60,
  "timestamp": ISODate("2025-03-10T09:45:00.000Z"),
  "expenses": 2500
},
{
  "_id": ObjectId("64a079cc4778b778187c2914"),
  "animal_name": "giraffe",
  "nature": "herbivore",
  "favorite_foods": [ "leaves", "shrubs" ],
  "care_taker_name": "Mike Johnson",
  "life_span_years": 20,
  "timestamp": ISODate("2025-04-05T14:20:00.000Z"),
  "expenses": 800
},
{
  "_id": ObjectId("64a079cc4778b778187c2915"),
  "animal_name": "bear",
  "nature": "herb",
  "favorite_foods": [ "fish", "berries" ],
  "care_taker_name": "Sarah Connor",
  "life_span_years": 35,
  "timestamp": ISODate("2025-05-12T16:00:00.000Z"),
  "expenses": 950
}
> animals.db.domestic_animals.find().pretty();
{
  "_id": ObjectId("64a079cc4778b778187c2916"),
  "animal_name": "dog",
  "nature": "herbivore",
  "favorite_foods": [ "meat", "vegetables" ],
  "care_taker_name": "John Doe",
  "life_span_years": 15,
  "timestamp": ISODate("2025-01-15T10:00:00.000Z"),
  "expenses": 1200
},
{
  "_id": ObjectId("64a079cc4778b778187c2917"),
  "animal_name": "cat",
  "nature": "herbivore",
  "favorite_foods": [ "fish", "birds" ],
  "care_taker_name": "John Smith",
  "life_span_years": 20,
  "timestamp": ISODate("2025-02-20T11:30:00.000Z"),
  "expenses": 1100
},
{
  "_id": ObjectId("64a079cc4778b778187c2918"),
  "animal_name": "parrot",
  "nature": "herbivore",
  "favorite_foods": [ "fruits", "seeds" ],
  "care_taker_name": "Mike Johnson",
  "life_span_years": 50,
  "timestamp": ISODate("2025-03-10T09:45:00.000Z"),
  "expenses": 2500
},
{
  "_id": ObjectId("64a079cc4778b778187c2919"),
  "animal_name": "hamster",
  "nature": "herbivore",
  "favorite_foods": [ "grains", "vegetables" ],
  "care_taker_name": "Mike Johnson",
  "life_span_years": 5,
  "timestamp": ISODate("2025-04-05T14:20:00.000Z"),
  "expenses": 800
},
{
  "_id": ObjectId("64a079cc4778b778187c291a"),
  "animal_name": "goldfish",
  "nature": "herbivore",
  "favorite_foods": [ "fish", "algae" ],
  "care_taker_name": "Sarah Connor",
  "life_span_years": 10,
  "timestamp": ISODate("2025-05-12T16:00:00.000Z"),
  "expenses": 950
}
```

Date: 26-02-2026

School: SCOPE

Reg. no.: 23BCE7208



Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

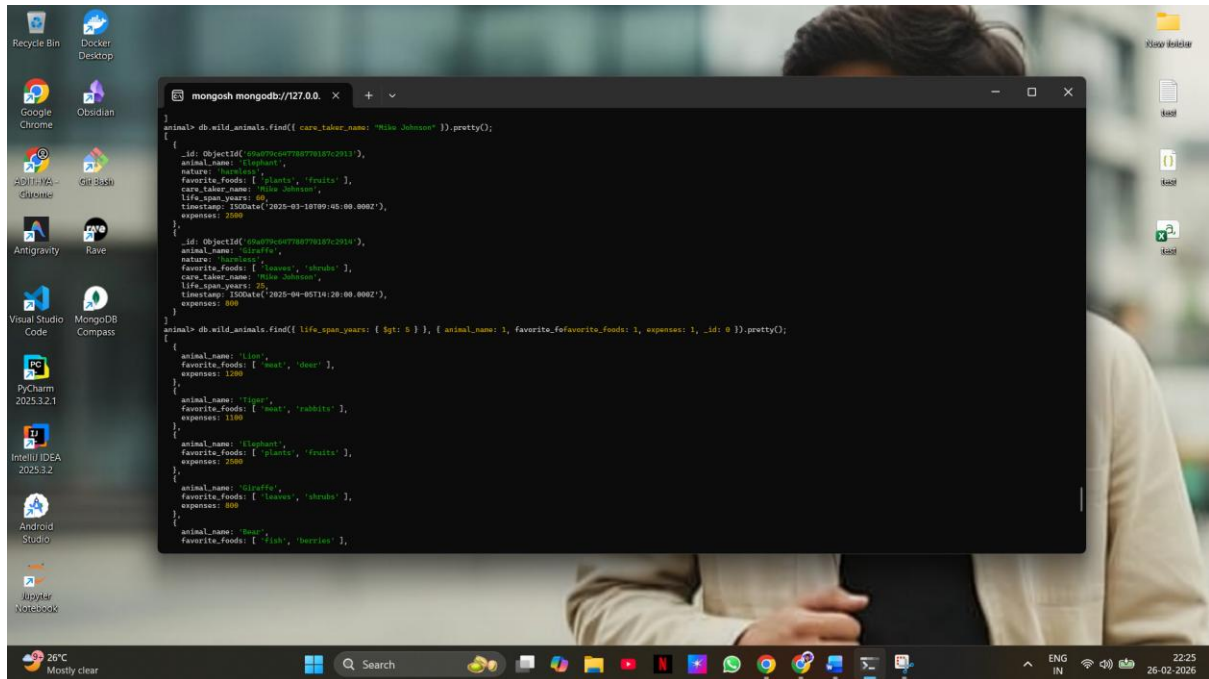
Faculty Name: Prof. S.Gopikrishnan

Student name: ADITHYA KUMAR

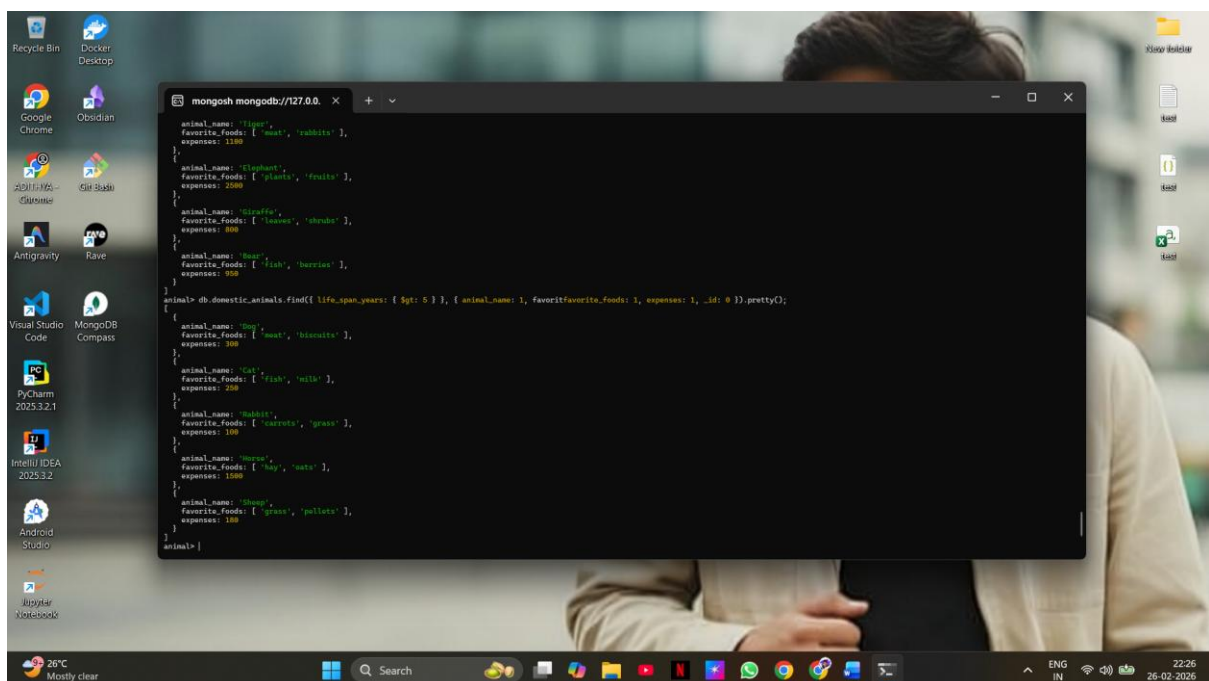
Date: 26-02-2026

School: SCOPE

Reg. no.: 23BCE7208



```
mongosh mongodb://127.0.0.1:27027
> use animals
> db.wild_animals.find({ caretaker_name: "Mike Johnson" }).pretty();
{
  "_id": ObjectId("49a070c4e7788779187c2013"),
  "animal_name": "Elephant",
  "nature": "herbivore",
  "favorite_foods": [ "plants", "fruits" ],
  "caretaker_name": "Mike Johnson",
  "life_span_years": 60,
  "timestamp": ISODate("2025-03-10T09:45:00.000Z"),
  "expenses": 2000
},
{
  "_id": ObjectId("49a070c4e7788779187c2014"),
  "animal_name": "Giraffe",
  "nature": "herbivore",
  "favorite_foods": [ "leaves", "shrubs" ],
  "caretaker_name": "Mike Johnson",
  "life_span_years": 25,
  "timestamp": ISODate("2025-04-05T14:20:00.000Z"),
  "expenses": 800
}
> db.wild_animals.find({ life_span_years: { $gt: 5 } }, { animal_name: 1, favorite_foods: 1, expenses: 1, _id: 0 }).pretty();
{
  "animal_name": "Lion",
  "favorite_foods": [ "meat", "deer" ],
  "expenses": 1200
},
{
  "animal_name": "Tiger",
  "favorite_foods": [ "meat", "rodents" ],
  "expenses": 1300
},
{
  "animal_name": "Elephant",
  "favorite_foods": [ "plants", "fruits" ],
  "expenses": 2000
},
{
  "animal_name": "Giraffe",
  "favorite_foods": [ "leaves", "shrubs" ],
  "expenses": 800
},
{
  "animal_name": "Bear",
  "favorite_foods": [ "fish", "berries" ],
  "expenses": 950
}
```



```
mongosh mongodb://127.0.0.1:27027
> use animals
> db.domestic_animals.find({ life_span_years: { $gt: 5 } }, { animal_name: 1, favorite_foods: 1, expenses: 1, _id: 0 }).pretty();
{
  "animal_name": "Dog",
  "favorite_foods": [ "meat", "biscuits" ],
  "expenses": 300
},
{
  "animal_name": "Cat",
  "favorite_foods": [ "fish", "milk" ],
  "expenses": 250
},
{
  "animal_name": "Rabbit",
  "favorite_foods": [ "carrots", "grass" ],
  "expenses": 100
},
{
  "animal_name": "Horse",
  "favorite_foods": [ "hay", "mats" ],
  "expenses": 1500
},
{
  "animal_name": "Sheep",
  "favorite_foods": [ "grass", "pellets" ],
  "expenses": 180
}
> animals
{
  "_id": ObjectId("49a070c4e7788779187c2013"),
  "animal_name": "Elephant",
  "nature": "herbivore",
  "favorite_foods": [ "plants", "fruits" ],
  "caretaker_name": "Mike Johnson",
  "life_span_years": 60,
  "timestamp": ISODate("2025-03-10T09:45:00.000Z"),
  "expenses": 2000
},
{
  "_id": ObjectId("49a070c4e7788779187c2014"),
  "animal_name": "Giraffe",
  "nature": "herbivore",
  "favorite_foods": [ "leaves", "shrubs" ],
  "caretaker_name": "Mike Johnson",
  "life_span_years": 25,
  "timestamp": ISODate("2025-04-05T14:20:00.000Z"),
  "expenses": 800
},
{
  "_id": ObjectId("49a070c4e7788779187c2015"),
  "animal_name": "Lion",
  "nature": "carnivore",
  "favorite_foods": [ "meat", "deer" ],
  "caretaker_name": "Mike Johnson",
  "life_span_years": 10,
  "timestamp": ISODate("2025-03-10T09:45:00.000Z"),
  "expenses": 1200
},
{
  "_id": ObjectId("49a070c4e7788779187c2016"),
  "animal_name": "Tiger",
  "nature": "carnivore",
  "favorite_foods": [ "meat", "rodents" ],
  "caretaker_name": "Mike Johnson",
  "life_span_years": 15,
  "timestamp": ISODate("2025-03-10T09:45:00.000Z"),
  "expenses": 1300
},
{
  "_id": ObjectId("49a070c4e7788779187c2017"),
  "animal_name": "Elephant",
  "nature": "herbivore",
  "favorite_foods": [ "plants", "fruits" ],
  "caretaker_name": "Mike Johnson",
  "life_span_years": 60,
  "timestamp": ISODate("2025-03-10T09:45:00.000Z"),
  "expenses": 2000
},
{
  "_id": ObjectId("49a070c4e7788779187c2018"),
  "animal_name": "Giraffe",
  "nature": "herbivore",
  "favorite_foods": [ "leaves", "shrubs" ],
  "caretaker_name": "Mike Johnson",
  "life_span_years": 25,
  "timestamp": ISODate("2025-04-05T14:20:00.000Z"),
  "expenses": 800
},
{
  "_id": ObjectId("49a070c4e7788779187c2019"),
  "animal_name": "Bear",
  "nature": "carnivore",
  "favorite_foods": [ "fish", "berries" ],
  "caretaker_name": "Mike Johnson",
  "life_span_years": 20,
  "timestamp": ISODate("2025-04-05T14:20:00.000Z"),
  "expenses": 950
},
{
  "_id": ObjectId("49a070c4e7788779187c2020"),
  "animal_name": "Dog",
  "nature": "carnivore",
  "favorite_foods": [ "meat", "biscuits" ],
  "caretaker_name": "Mike Johnson",
  "life_span_years": 10,
  "timestamp": ISODate("2025-03-10T09:45:00.000Z"),
  "expenses": 300
},
{
  "_id": ObjectId("49a070c4e7788779187c2021"),
  "animal_name": "Cat",
  "nature": "carnivore",
  "favorite_foods": [ "fish", "milk" ],
  "caretaker_name": "Mike Johnson",
  "life_span_years": 15,
  "timestamp": ISODate("2025-03-10T09:45:00.000Z"),
  "expenses": 250
},
{
  "_id": ObjectId("49a070c4e7788779187c2022"),
  "animal_name": "Rabbit",
  "nature": "herbivore",
  "favorite_foods": [ "carrots", "grass" ],
  "caretaker_name": "Mike Johnson",
  "life_span_years": 5,
  "timestamp": ISODate("2025-03-10T09:45:00.000Z"),
  "expenses": 100
},
{
  "_id": ObjectId("49a070c4e7788779187c2023"),
  "animal_name": "Horse",
  "nature": "herbivore",
  "favorite_foods": [ "hay", "mats" ],
  "caretaker_name": "Mike Johnson",
  "life_span_years": 20,
  "timestamp": ISODate("2025-03-10T09:45:00.000Z"),
  "expenses": 1500
},
{
  "_id": ObjectId("49a070c4e7788779187c2024"),
  "animal_name": "Sheep",
  "nature": "herbivore",
  "favorite_foods": [ "grass", "pellets" ],
  "caretaker_name": "Mike Johnson",
  "life_span_years": 10,
  "timestamp": ISODate("2025-03-10T09:45:00.000Z"),
  "expenses": 180
}
> animals
{
  "_id": ObjectId("49a070c4e7788779187c2013"),
  "animal_name": "Elephant",
  "nature": "herbivore",
  "favorite_foods": [ "plants", "fruits" ],
  "caretaker_name": "Mike Johnson",
  "life_span_years": 60,
  "timestamp": ISODate("2025-03-10T09:45:00.000Z"),
  "expenses": 2000
},
{
  "_id": ObjectId("49a070c4e7788779187c2014"),
  "animal_name": "Giraffe",
  "nature": "herbivore",
  "favorite_foods": [ "leaves", "shrubs" ],
  "caretaker_name": "Mike Johnson",
  "life_span_years": 25,
  "timestamp": ISODate("2025-04-05T14:20:00.000Z"),
  "expenses": 800
},
{
  "_id": ObjectId("49a070c4e7788779187c2015"),
  "animal_name": "Lion",
  "nature": "carnivore",
  "favorite_foods": [ "meat", "deer" ],
  "caretaker_name": "Mike Johnson",
  "life_span_years": 10,
  "timestamp": ISODate("2025-03-10T09:45:00.000Z"),
  "expenses": 1200
},
{
  "_id": ObjectId("49a070c4e7788779187c2016"),
  "animal_name": "Tiger",
  "nature": "carnivore",
  "favorite_foods": [ "meat", "rodents" ],
  "caretaker_name": "Mike Johnson",
  "life_span_years": 15,
  "timestamp": ISODate("2025-03-10T09:45:00.000Z"),
  "expenses": 1300
},
{
  "_id": ObjectId("49a070c4e7788779187c2017"),
  "animal_name": "Elephant",
  "nature": "herbivore",
  "favorite_foods": [ "plants", "fruits" ],
  "caretaker_name": "Mike Johnson",
  "life_span_years": 60,
  "timestamp": ISODate("2025-03-10T09:45:00.000Z"),
  "expenses": 2000
},
{
  "_id": ObjectId("49a070c4e7788779187c2018"),
  "animal_name": "Giraffe",
  "nature": "herbivore",
  "favorite_foods": [ "leaves", "shrubs" ],
  "caretaker_name": "Mike Johnson",
  "life_span_years": 25,
  "timestamp": ISODate("2025-04-05T14:20:00.000Z"),
  "expenses": 800
},
{
  "_id": ObjectId("49a070c4e7788779187c2019"),
  "animal_name": "Bear",
  "nature": "carnivore",
  "favorite_foods": [ "fish", "berries" ],
  "caretaker_name": "Mike Johnson",
  "life_span_years": 20,
  "timestamp": ISODate("2025-04-05T14:20:00.000Z"),
  "expenses": 950
},
{
  "_id": ObjectId("49a070c4e7788779187c2020"),
  "animal_name": "Dog",
  "nature": "carnivore",
  "favorite_foods": [ "meat", "biscuits" ],
  "caretaker_name": "Mike Johnson",
  "life_span_years": 10,
  "timestamp": ISODate("2025-03-10T09:45:00.000Z"),
  "expenses": 300
},
{
  "_id": ObjectId("49a070c4e7788779187c2021"),
  "animal_name": "Cat",
  "nature": "carnivore",
  "favorite_foods": [ "fish", "milk" ],
  "caretaker_name": "Mike Johnson",
  "life_span_years": 15,
  "timestamp": ISODate("2025-03-10T09:45:00.000Z"),
  "expenses": 250
},
{
  "_id": ObjectId("49a070c4e7788779187c2022"),
  "animal_name": "Rabbit",
  "nature": "herbivore",
  "favorite_foods": [ "carrots", "grass" ],
  "caretaker_name": "Mike Johnson",
  "life_span_years": 5,
  "timestamp": ISODate("2025-03-10T09:45:00.000Z"),
  "expenses": 100
},
{
  "_id": ObjectId("49a070c4e7788779187c2023"),
  "animal_name": "Horse",
  "nature": "herbivore",
  "favorite_foods": [ "hay", "mats" ],
  "caretaker_name": "Mike Johnson",
  "life_span_years": 20,
  "timestamp": ISODate("2025-03-10T09:45:00.000Z"),
  "expenses": 1500
},
{
  "_id": ObjectId("49a070c4e7788779187c2024"),
  "animal_name": "Sheep",
  "nature": "herbivore",
  "favorite_foods": [ "grass", "pellets" ],
  "caretaker_name": "Mike Johnson",
  "life_span_years": 10,
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  "expenses": 180
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